# 3008ICT Deep Learning

## Week 2 Part 2. Softmax Regression/Classification

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## Adapted from slides at d2l.ai

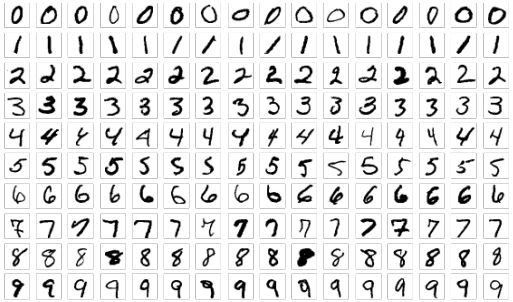
# Outline

- (Non-Linear) Regression
- Regression vs Classification
- Multiclass Classification
- Softmax and Cross Entropy Loss
- Loss Functions

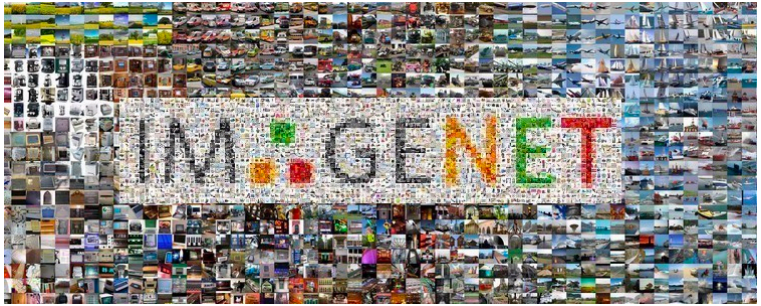
# Regression vs. Multi-class Classification

- Regression estimates a continuous value
- Classification predicts a discrete category

MNIST: classify hand-written digits (10 classes)

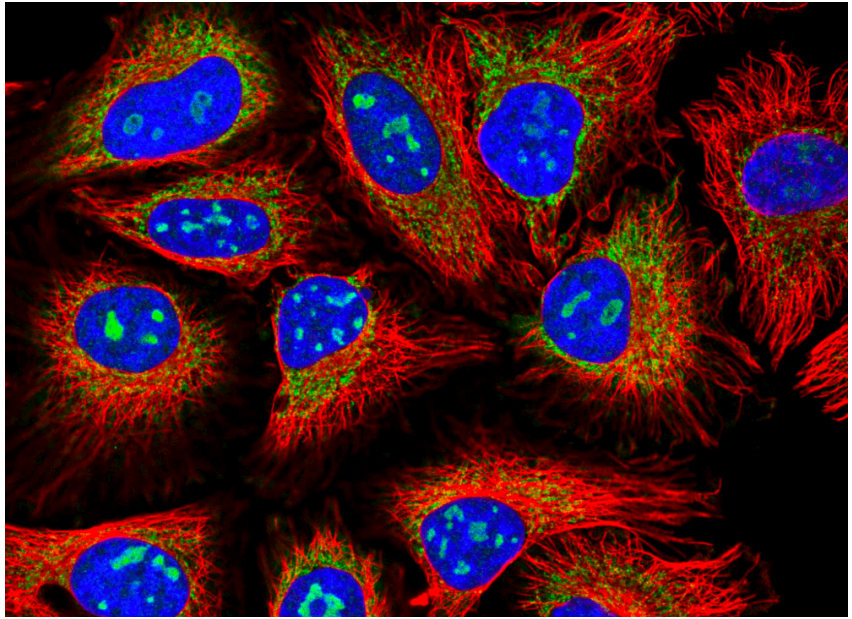


ImageNet: classify nature objects (1000 classes)



# Classification Tasks at Kaggle

Classify human protein microscope images into 28 categories



```
0.  Nucleoplasm
1.  Nuclear membrane
2.  Nucleoli
3.  Nucleoli fibrillar
4.  Nuclear speckles
5.  Nuclear bodies
6.  Endoplasmic reticu
7.  Golgi apparatus
8.  Peroxisomes
9.  Endosomes
10.  Lysosomes
11.  Intermediate fila
12.  Actin filaments
13.  Focal adhesion si
14.  Microtubules
15.  Microtubule ends
16.  Cytokinetic bridg
```

https://www.kaggle.com/c/human-protein-atlas-image-classification

# Classification Tasks at Kaggle

Classify malware into 9 categories

https://www.kaggle.com/c/malware-classification

# Classification Tasks at Kaggle

Classify toxic Wikipedia comments into 7 categories

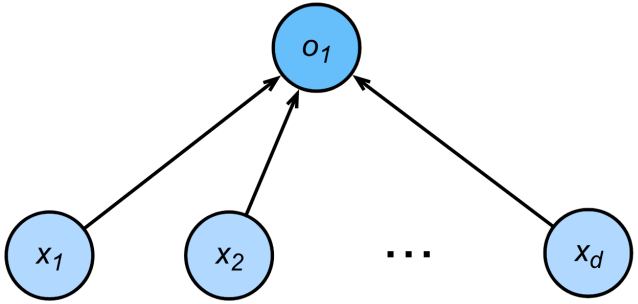
| comment_text | toxic | severe_toxic | obsc |
|---|---|---|---|
| Explanation\nWhy the edits made under my usern... | 0 | 0 | 0 |
| D'aww! He matches this background colour I'm s... | 0 | 0 | 0 |
| Hey man, I'm really not trying to edit war. It... | 0 | 0 | 0 |
| "\nMore\nI can't make any real suggestions on ... | 0 | 0 | 0 |
| You, sir, are my hero. Any chance you remember... | 0 | 0 | 0 |

https://www.kaggle.com/c/jigsaw-toxic-comment-classification-challenge

# From Regression to Multi-class Classification

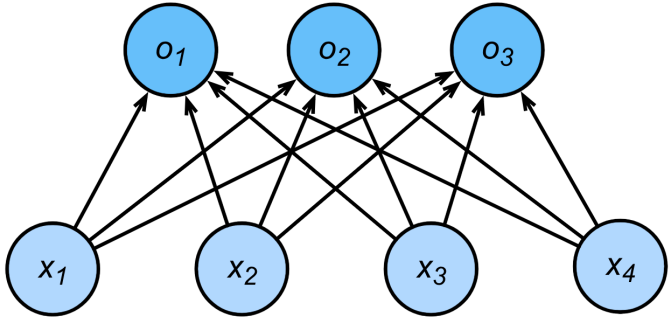
## Regression

- Single continuous output
- Natural scale in $\mathbb{R}$
- Loss given e.g. in terms of difference $y - f(x)$




## Classification

- Multiple classes, typically multiple outputs
- Score *should* reflect confidence …



# From Regression to Multi-class Classification

## Square Loss

- One hot encoding per class

$$\mathbf{y} = [y_1, y_2, \ldots, y_n]^\top$$

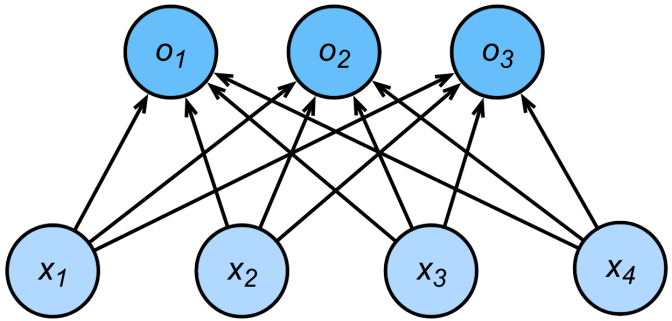
$$y_i = \begin{cases} 1 \text{ if } i = y \\ 0 \text{ otherwise} \end{cases}$$

- Train with squared loss
- Largest output wins

$$\hat{y} = \operatorname*{argmax}_i o_i$$

## Classification

- Multiple classes, typically multiple outputs
- Score *should* reflect confidence …



# From Regression to Multi-class Classification

## Uncalibrated Scale

- One output per class
- Largest output wins

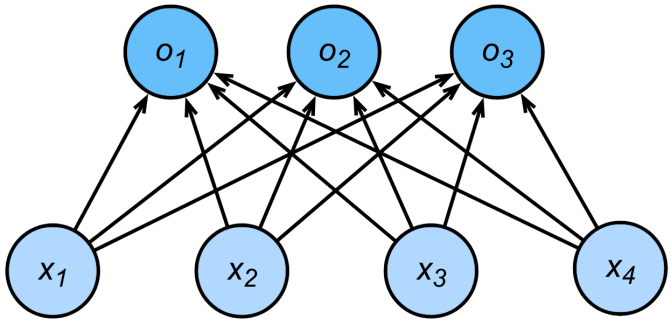
$$\hat{y} = \operatorname*{argmax}_i o_i$$

- Want that correct class is recognized confidently (**large margin**)

$$o_y - o_i \geq \Delta(y, i)$$

## Classification

- Multiple classes, typically multiple outputs
- Score *should* reflect confidence …



# From Regression to Multi-class Classification

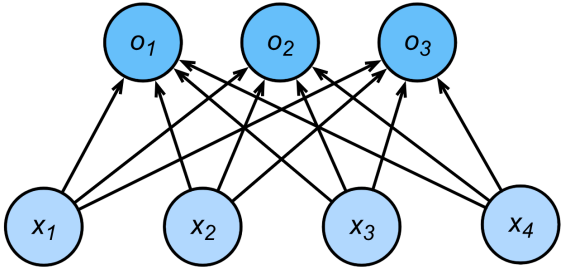
## Calibrated Scale

- Output matches probabilities (nonnegative, sums to 1)
- $$\mathrm{softmax}(\hat{y}_i) = \frac{e^{o_i}}{\sum_{j=1}^{b} e^{o_j}}$$
- $$\mathrm{softmax}(\hat{y}) = (\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_m)^\top$$

Here $m$ is the number of output nodes.

## Classification

- Multiple classes, typically multiple outputs
- Score *should* reflect confidence …



# Softmax and Cross-Entropy Loss

- We could use the difference $y - \hat{y}$ between $y$ and $\hat{y}$ as the loss, but the cross entropy is better.
- Cross-Entropy Loss (for probability distribution y)

$$l(y, \hat{y}) = -\sum_i y_i \, log \, \hat{y}_i$$

- **Gradient**

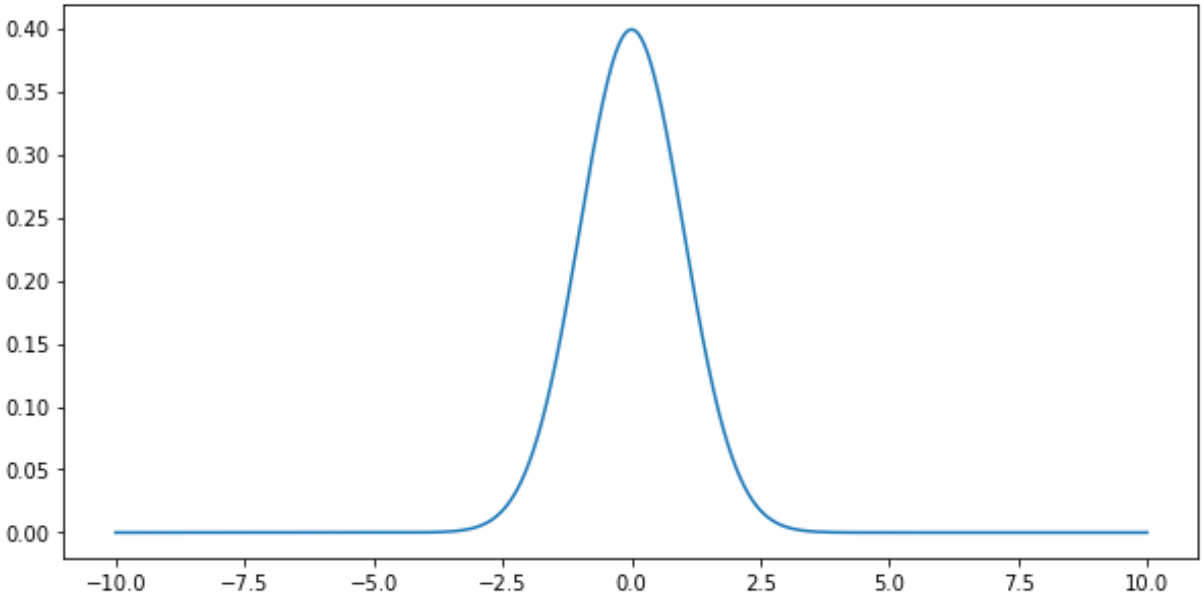
Difference between true and estimated probability

$$\partial_{o_i} l(y, \hat{y}) = \mathrm{softmax}(\mathrm{O_i}) - y_i$$

# Flashback - Normal Distribution

Density

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$




$\mu$: mean

$\sigma$: deviation

# Maximum likelihood estimation

- Data - ‘student didn’t do homework’
- Possible parameters
  - **‘dog ate homework’**
  - **‘abducted by aliens’**
  - **‘too lazy’**
  - **‘sick grandmother’**
- **All parameters explain the data.**
  - p(‘no homework’|explanation) = 1 (all explanations work)

Should we believe all?

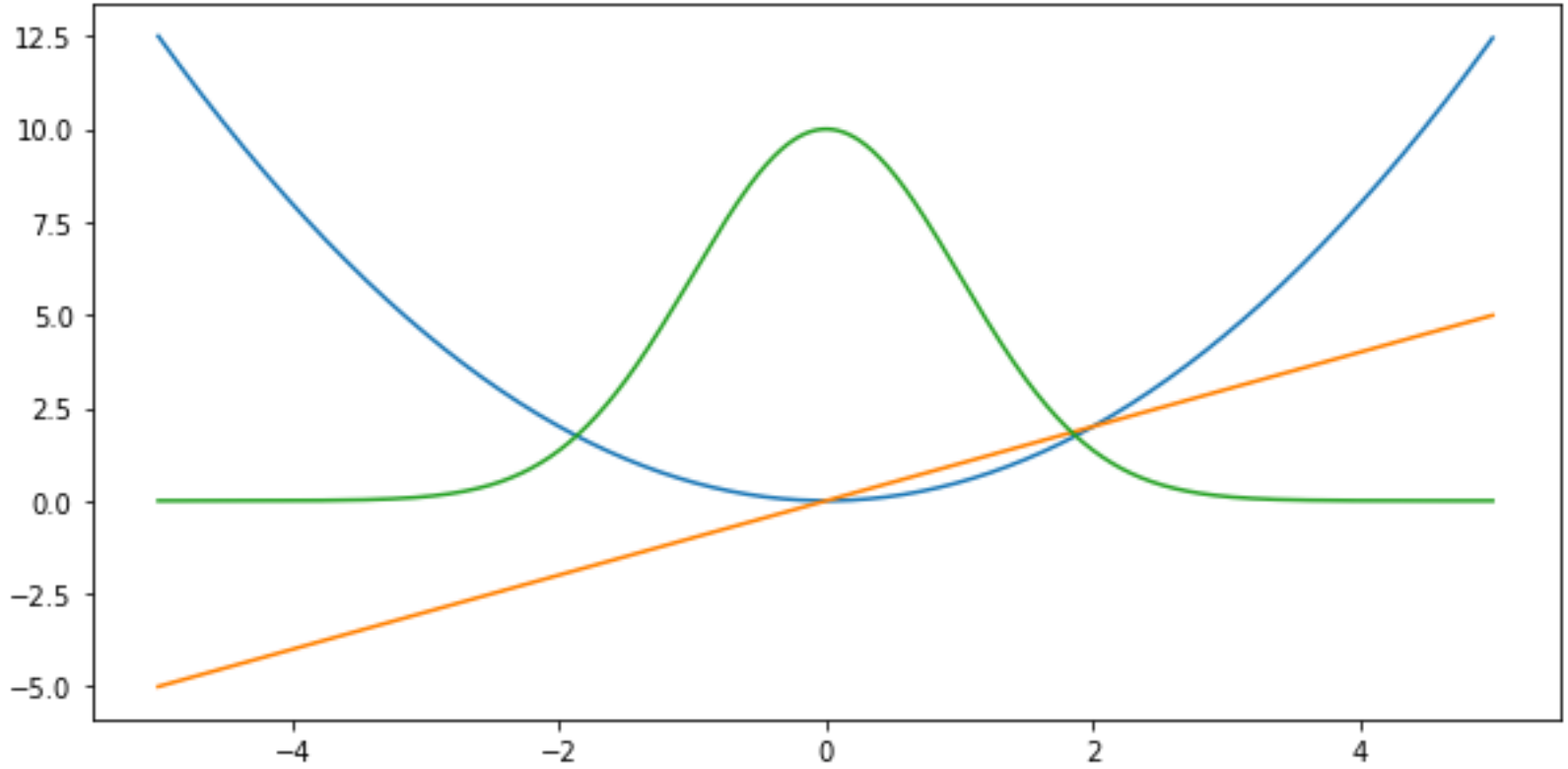
| lazy student | grandma sick | dog ate it | alien abduction |
|---|---|---|---|
| **0.8** | 0.19 | 0.0099 | 0.0001 |

MAGIC Etch A Sketch® SCREEN
Loss
Functions
Horizontal Dial
OHIO ART
The World of Toys®
Vertical Dial
MAGIC SCREEN IS GLASS SET IN STURDY PLASTIC FRAME
USE WITH CARE

# L2 Loss

$$l(y, y') = \frac{1}{2}(y - y')^2$$

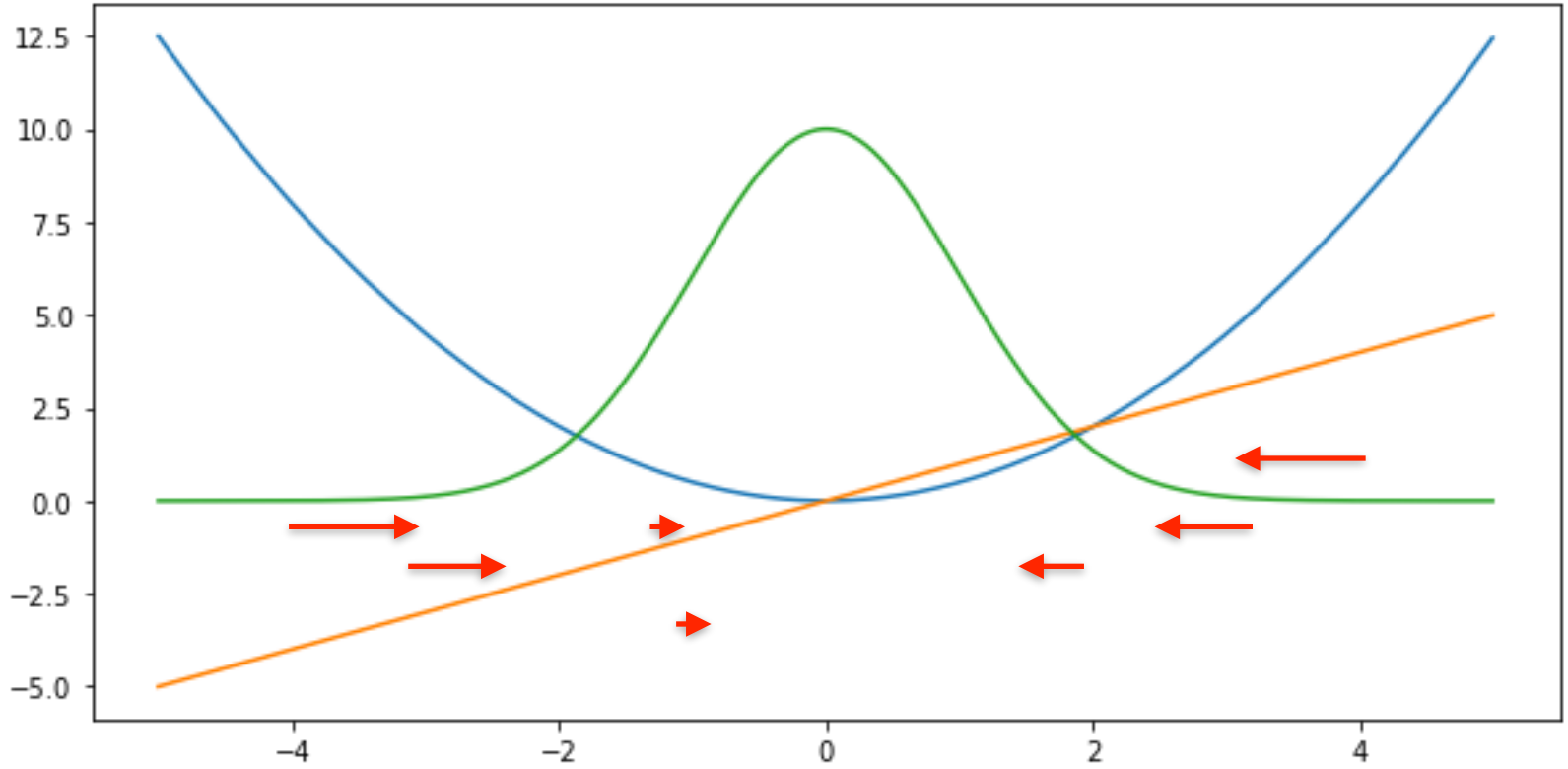
In:



# L2 Loss - mean

$$l(y, y') = \frac{1}{2}(y - y')^2$$

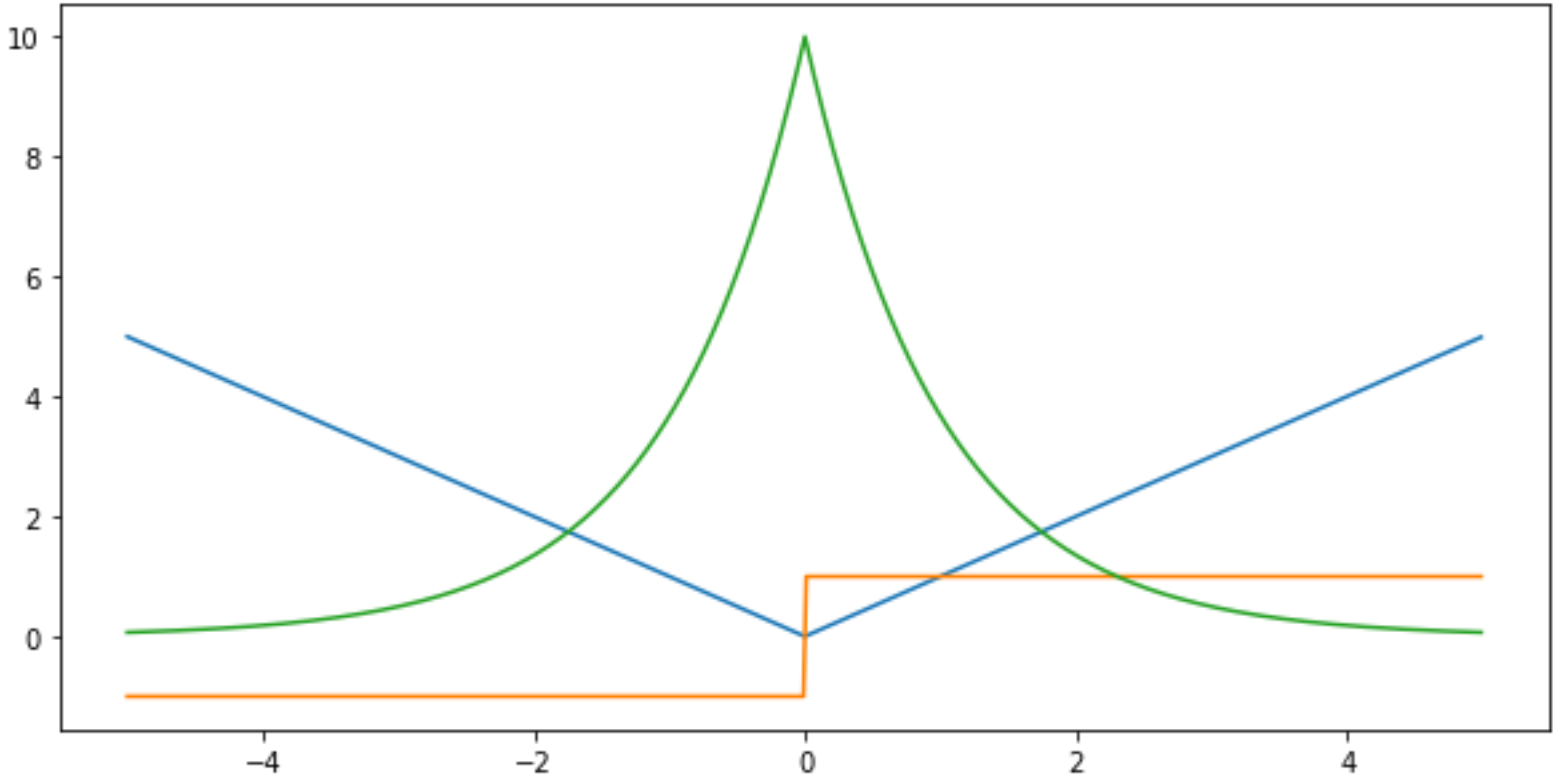
In:



# L1 Loss

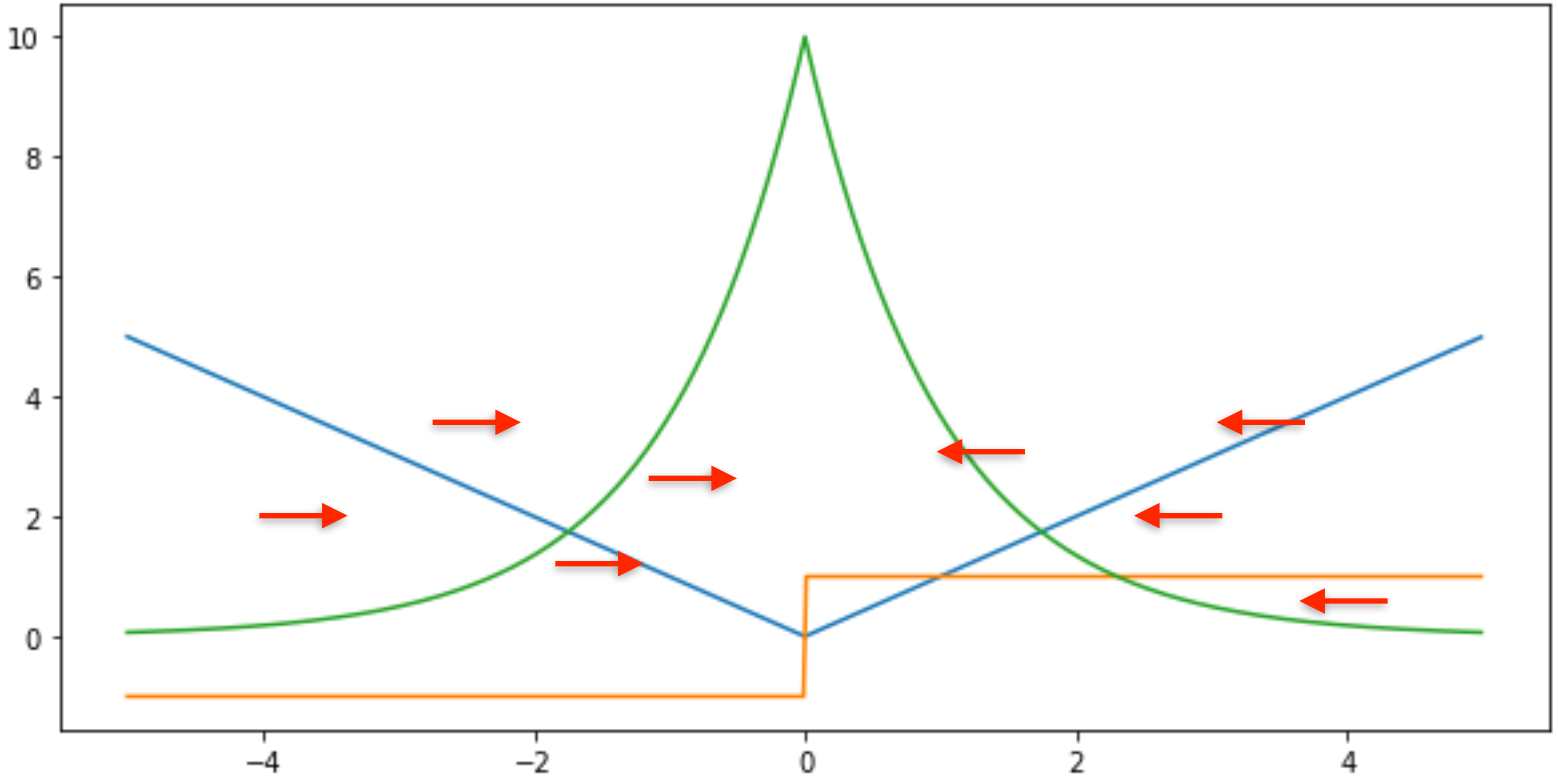
$$l(y, y') = |y - y'|$$

In:



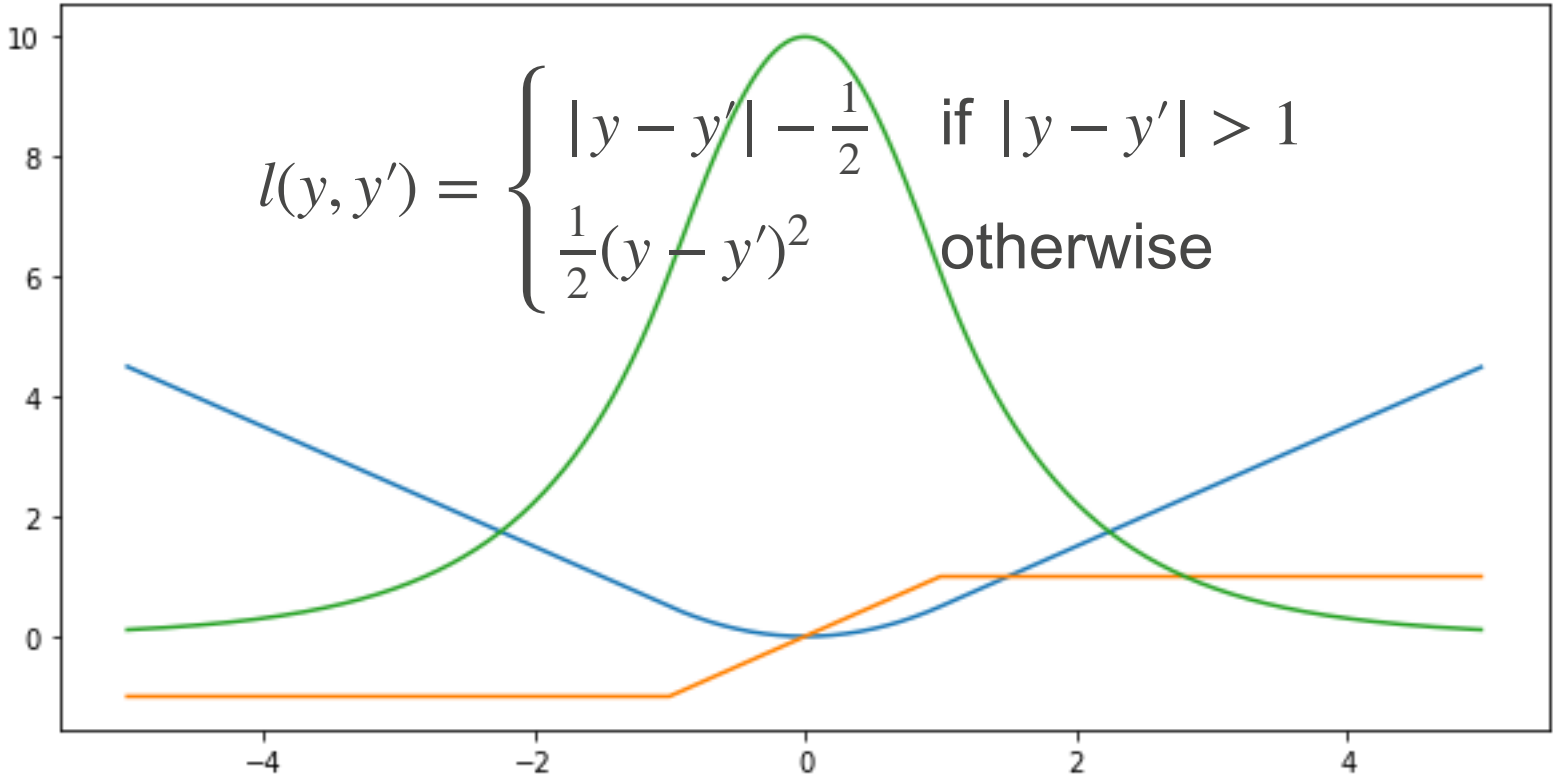
# L1 Loss - median $l(y, y') = |y - y'|$

In:



# Huber's Robust Loss

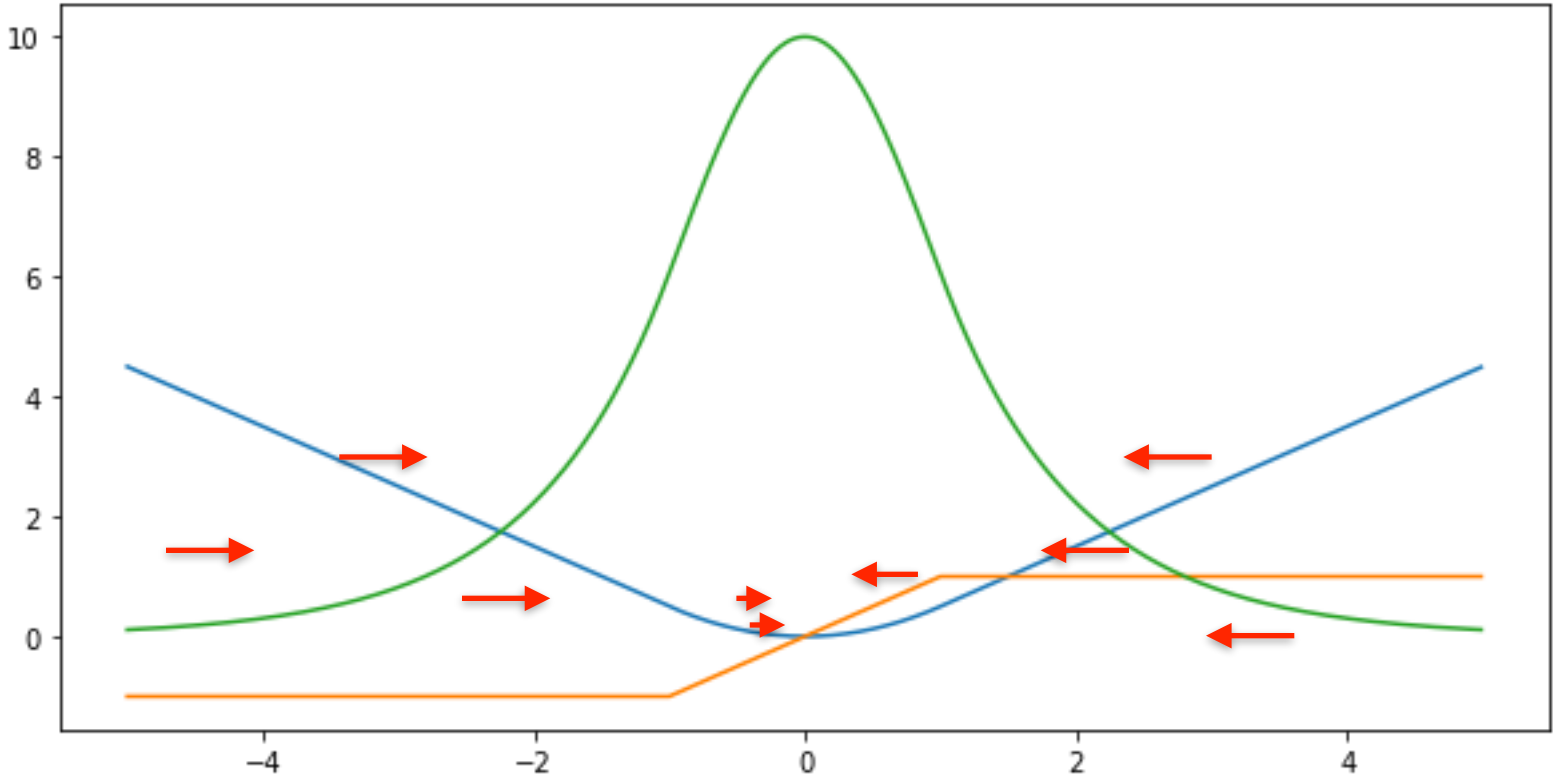
In:




$$l(y, y') = \begin{cases} |y - y'| - \frac{1}{2} & \text{if } |y - y'| > 1 \\ \frac{1}{2}(y - y')^2 & \text{otherwise} \end{cases}$$

# Huber’s Robust Loss - trimmed mean

In:



# Datasets for Image Classification

| Dataset | Training Images | Test Images | Classes | Resolution | Color Channels |
|---|---|---|---|---|---|
| **MNIST** | 60,000 | 10,000 | 10 | $28 \times 28$ | 1 (Grayscale |
| **Fashion-MNIST** | 60,000 | 10,000 | 10 | $28 \times 28$ | 1 (Grayscale |
| **ImageNet-1K** | ~1,280,000 | 100,000 | 1,000 | $\sim 482 \times 415$ (avg) | 3 (RGB) |

# Code

# Summary

- **Softmax Regression is a model for multiple-class classification**
- **Softmax operator can be used to obtain the pre-confidence of each class (i.e., probability)**
- **Cross entropy is used to measure the loss (i.e., difference between predicted class and truth class)**